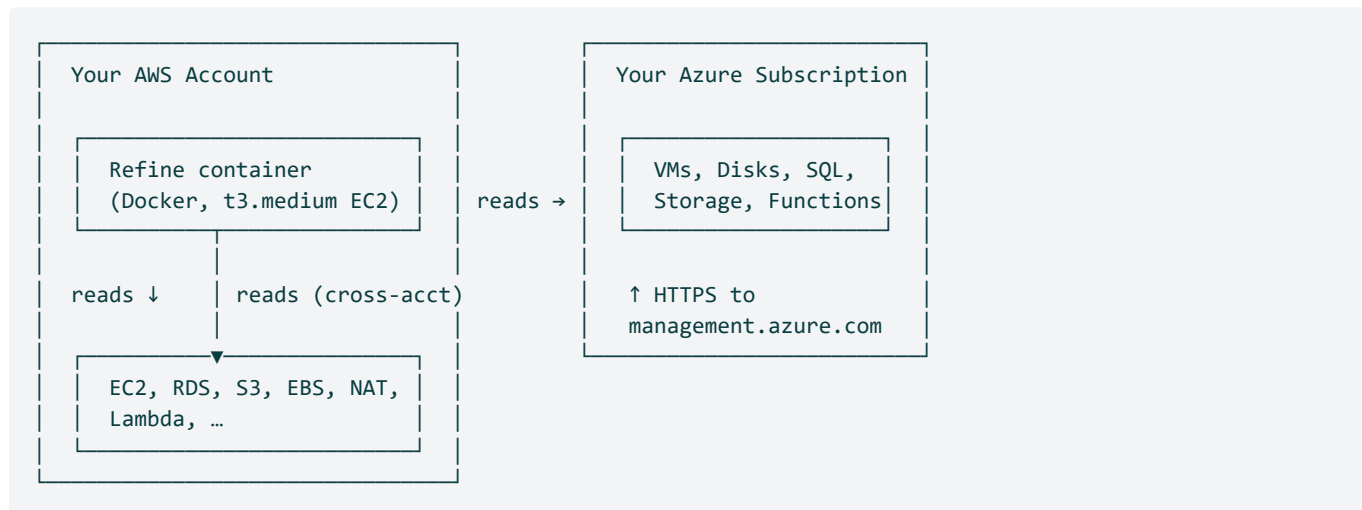


Refine — Customer Quick-Start

Single end-to-end guide: set up the Refine server on AWS, connect your AWS and Azure accounts, learn the daily-use flow, and troubleshoot common issues.

If you only read one page, read this one. Everything else in the `docs/` folder is reference material.

What you're about to set up



You'll deploy:

1. **The Refine server** — a single Docker container on an AWS EC2 instance that you own. Uses a SQLite database; no cloud database to manage.
2. **A CloudFormation stack in your AWS account** — gives Refine read-only permissions to your AWS resources. Refine never gets write access.
3. **A Service Principal in your Azure tenant** — gives Refine read-only access to Azure resources. Same model: read-only.

Total time, end to end: **~60 minutes**.

How did you get Refine? (purchase paths)

Refine ships through a few channels with slightly different setup flows. Pick the one that matches how your subscription was created:

Channel	What it looks like	Where to go
Public AWS Marketplace (annual contract)	You subscribed to <i>Blacktip Refine — Multi-Cloud Cost Intelligence</i> on AWS Marketplace by choosing a paid tier (Starter / Core / Growth / Scale), then launched a CloudFormation stack from the listing. No license file and no activation code — Refine auto-verifies your Marketplace contract entitlement through AWS License Manager on boot.	Stop here. Follow docs/marketplace/DEPLOYMENT_GUIDE.md instead — the Marketplace flow handles EC2 provisioning, image pull, the entitlement check, and per-account onboarding through a simpler path. Then come back here for Part 2 (connecting Azure / GCP) and Part 3 (daily-use flow).
Free edition (\$0)	You deployed Blacktip Refine Free — \$0, self-hosted, up to \$1,000/month of analyzed cloud spend, funded by an anonymous aggregated monthly telemetry summary (total spend + savings only). It's a template-asserted Free entitlement (no License Manager contract) and pauses with an upgrade prompt over the cap.	Deploy via the same AWS Marketplace CloudFormation stack — follow docs/marketplace/DEPLOYMENT_GUIDE.md , then return here for Part 2 (Azure / GCP) and Part 3 (daily-use flow).
Direct sales / GovCloud (BYOL)	Blacktip Solutions sent you a delivery ZIP (~1.5 GB) via email, S3 link, or encrypted USB containing <code>refine.license</code> , <code>infra/refine-server-auto.yaml</code> , the Docker image tarball (for GovCloud / air-gap), and day-2 management scripts — plus a separate activation code. (Private Offers are <i>not</i> here — they run through AWS Marketplace as contract entitlements; use the Public Marketplace row above.)	Continue with this guide.

This guide covers the **direct-sales / BYOL path** end-to-end. If you're on the public Marketplace contract path, the deployment steps in Part 1 are handled for you by the Marketplace CloudFormation; skip to **Part 2** below once your Refine UI loads.

Before you begin — what you'll need

Item	Notes
AWS account with admin permissions	To launch the EC2 host stack and the per-account onboarding stack
Azure subscription with Owner or User Access Administrator	To create the Service Principal
Your Refine delivery ZIP from Blacktip (direct-sales / BYOL)	Contains <code>refine.license</code> , setup wizard, and config. Public Marketplace subscribers don't get a ZIP — skip to Part 2.
Your <code>refine.license</code> file	Inside the ZIP. Bound to your installation. (Public Marketplace uses an auto-verified contract entitlement instead — no file.)
An email address that gets used for Refine login	The first user you create is the root admin
30 GB free disk space on the EC2 host	For container + database + cost data
Outbound HTTPS access from the EC2 host	To <code>management.azure.com</code> , <code>login.microsoftonline.com</code> , and S3 endpoints

If you're in a regulated industry (GovCloud, FedRAMP), stop here and read `AIR_GAPPED_SETUP_GUIDE.md` instead.

Part 1 — Deploy the Refine server on AWS EC2

Direct-sales / BYOL path only. These deployment steps apply if Blacktip sent you a delivery ZIP. Public AWS Marketplace (contract) subscribers skip Part 1 entirely — the Marketplace CloudFormation product provisions the EC2, pulls the image from AWS Public ECR, and auto-verifies the contract entitlement through AWS License Manager. Jump to [Part 2](#) once your Refine UI loads.

The Refine server runs as a single Docker container. The fastest path is to launch it from a CloudFormation template that creates the EC2, IAM role, security group, and Elastic IP for you.

1.1 — Upload the delivery ZIP to S3

The CloudFormation template downloads the Refine delivery ZIP from an S3 bucket you control. So first:

```
# Create a private S3 bucket (or reuse one)
aws s3 mb s3://my-refine-delivery-bucket --region us-east-1

# Upload the ZIP Blacktip sent you
aws s3 cp refine-CUSTOMER-vX.Y.Z.zip s3://my-refine-delivery-bucket/refine-delivery.zip
```

The S3 object name **must be exactly** `refine-delivery.zip` — the CloudFormation template hardcodes that name.

1.2 — Launch the host CloudFormation stack

In the AWS Console:

1. **CloudFormation** → **Create stack** → **With new resources**
2. **Upload a template file** → pick `infra/refine-server-auto.yaml` from your delivery ZIP
3. Click **Next** and fill in parameters:

Parameter	Value	Notes
StackName	<code>refine-server</code>	Anything you like
InstanceType	<code>t3.medium</code>	4 GB RAM. <code>t3.small</code> (2 GB) works for ≤ 100 resources; <code>t3.micro</code> (1 GB) will OOM
DiskSizeGB	<code>30</code>	30 is plenty for a year of data
AllowedAppCidr	<code>1.2.3.4/32</code>	Your office / VPN egress IP. See "How to find your CIDR" below. Use <code>0.0.0.0/0</code> only if you'll put Refine behind an ALB / Cloudflare.
EnableHttps	<code>false</code> for now	Flip to <code>true</code> once you have a domain + ACM cert
UseExistingBucket	<code>true</code>	You created and pre-loaded one in step 1.1
DeliveryBucketName	<code>my-refine-delivery-bucket</code>	The bucket from 1.1
KeyPairName	(your EC2 key pair)	For SSH if needed; otherwise leave blank to use SSM Session Manager

How to find your `AllowedAppCidr` — this is the public IP address you'll connect to Refine from. Pick the path that matches where you'll use Refine:

- **From a single laptop / home office:** open a new browser tab and go to <https://whatismyip.com> or <https://ifconfig.me>. Copy the IPv4 address shown (e.g. `203.0.113.42`) and add `/32` to the end → `203.0.113.42/32`. The `/32` means "this exact IP, nothing else."
- **From an office network (multiple people, one egress):** ask your IT team for the office's **public egress IP range** in CIDR notation. It usually looks like `203.0.113.0/24` (256 IPs) or `203.0.113.0/29` (8 IPs). If they're not sure, every person on the network sees the same IP at whatismyip.com — use that with `/32` to start, then expand if needed.
- **From a corporate VPN:** find your VPN's egress IP. While connected to the VPN, go to whatismyip.com. That's the address you need. Add `/32`. If your team uses multiple VPN exit nodes, ask IT for the full CIDR range.
- **From AWS (e.g. a bastion host):** use the bastion's Elastic IP `/32`.
- **You'll connect from multiple places (home + office + on the road):** you can list multiple CIDRs separated by commas in some CF templates, but the simplest path is to put Refine behind an Application Load Balancer with `0.0.0.0/0` inbound on the EC2 SG and lock down the ALB with your SSO. That's a Day-2 upgrade — for the first install, just use one `/32` and adjust later.

Your IP can change. Home ISPs and small offices often get a new public IP every few weeks. If you suddenly can't reach Refine, re-check whatismyip.com and update the security group rule. See the Troubleshooting matrix in Part 6 for the exact AWS CLI command.

4. Acknowledge IAM creation → **Create stack**.
5. Wait ~5 minutes for `CREATE_COMPLETE`. Then **Outputs** → copy the **Elastic IP**.

1.3 — Connect to the EC2 (SSM or SSH)

The CloudFormation template launched the EC2 and ran a user-data script that installed Docker and extracted your delivery ZIP to `/opt/refine`. **You now need to log into the EC2 to finish the install** (run the setup wizard, which creates `.env`, logs Docker into the registry, pulls the image, and starts the container).

Two scripts, two different jobs — read this once:

Script	When to run	What it does
<code>setup.sh</code>	First time only. First thing you run after logging in.	Checks prerequisites, copies <code>.env.example</code> → <code>.env</code> , logs Docker into the <code>ghcr.io</code> registry with your bundled token, pulls the image, starts the container.
<code>refine.sh</code>	Day-2 ops, after <code>setup.sh</code> finished.	Status checks, start/stop, view logs, pull updates, take backups. Don't run this first — it expects the install to already be done.

Option A — SSM Session Manager (recommended, no SSH key needed)

- **AWS Console:** EC2 → Instances → select `refine-server` → **Connect** → **Session Manager** tab → **Connect**
- **CLI:** `aws ssm start-session --target i-XXXXXXXXXXXX` (instance ID is in the CloudFormation stack **Outputs** tab, key `InstanceId`)

You'll land in a shell as `ssm-user` in `/`. Continue to step 1.4 below.

Option B — SSH (if you provided a `KeyPairName` at stack creation)

```
ssh -i ~/.ssh/your-key.pem ec2-user@<your-elastic-ip>
# (Use 'ubuntu@' instead of 'ec2-user@' if the AMI is Ubuntu)
```

You'll land in a shell as `ec2-user` (or `ubuntu`). Continue to step 1.4 below.

1.4 — Run the setup wizard (`setup.sh`)

You're now logged in. Run these commands **in order, exactly as shown**:

```
# 1. Become root (needed for Docker + writing into /opt/refine)
sudo su -

# 2. Move to the install directory
cd /opt/refine

# 3. Confirm the install files are in place
ls
# You should see at minimum:
#   setup.sh  setup.py  refine.sh  docker-compose.yml
#   .env.example  refine.license  registry-login.sh  scripts/

# 4. Run the install wizard
./setup.sh
```

`setup.sh` runs for ~1–2 minutes (mostly pulling the Docker image). When it finishes you'll see a success banner with the Refine URL. The container is now running.

Troubleshooting `setup.sh`:

- `setup.sh: command not found` → run `bash setup.sh` instead (file may have lost `+x` on extract)
- `Permission denied` → confirm you ran `sudo su -` first
- `Compose file not found` → confirm you're in `/opt/refine` (run `pwd`). Also confirm the file is named `docker-compose.yml` and not `docker-compose.customer.yml` (older deliveries; if so, rename it: `mv docker-compose.customer.yml docker-compose.yml`)
- **Fully-manual fallback** if `setup.sh` keeps erroring:

```
cp .env.example .env
bash registry-login.sh      # Logs Docker into ghcr.io with bundled token
docker compose pull
docker compose up -d
docker compose ps           # should show "refine" Up (healthy)
```

1.5 — Open Refine in your browser

Back on your laptop, open:

```
http://<your-elastic-ip>:8000
```

You should see the **Welcome to Refine** screen. Click through:

1. **Create your first user** — email + password. This is the root admin.
2. **License check** — Refine reads `/app/refine.license` automatically. Confirm it shows your customer name and expiration date.
3. **Done.** You're at the Dashboard. It's empty — that's expected; we haven't connected any cloud accounts yet.

1.6 — Verify health (and day-2 ops cheat sheet)

Back in the SSM/SSH shell, the `refine.sh` script is your day-2 toolkit:

```
./refine.sh status      # should show refine "Up (healthy)"
./refine.sh health     # prints {"status":"ok",...} from /health
./refine.sh url        # prints the URL to open
./refine.sh version    # prints running Refine version
./refine.sh logs -f    # tail logs live (Ctrl+C to stop)
./refine.sh restart    # restart the container
./refine.sh update     # pull latest image + recreate
./refine.sh backup     # snapshot data dir to backups/
```

If `./refine.sh status` shows nothing running, the container has stopped:

```
tail -80 /var/log/cloud-init-output.log # first-boot user-data log
./refine.sh start
./refine.sh logs -f
```

If any step in 1.4 / 1.5 / 1.6 still fails, jump to [Part 6 — Troubleshooting](#).

Part 2 — Connect your AWS account

This deploys a CloudFormation stack in **the AWS account you want Refine to monitor**. That can be the same account that hosts Refine, a different account, or both. Repeat for every account you want monitored.

2.1 — Start the Add Account flow

In Refine: **Cloud Accounts** → **Add Account** → **AWS**.

Fill in:

Field	Value
AWS Account ID	The 12-digit ID of the account to monitor
Deployment Region	The region where you'll create the CloudFormation stack. Pick somewhere you have most of your resources (e.g. <code>us-east-1</code>)
Account Name	Optional — friendly label like "Production"

Click **Add Account**. Refine generates an **External ID** unique to this account link and presents the **Account Setup** modal.

2.2 — Download and deploy the CloudFormation template

In the Account Setup modal:

1. Pick the same region you chose in 2.1
2. Click **Download CloudFormation Template** — the External ID is baked in; don't edit the file
3. In the AWS Console for the target account: **CloudFormation** → **Create stack** → **Upload template**
4. Accept all defaults — the template is parameterless (it reads the External ID from inside)
5. Acknowledge IAM creation → **Create stack**
6. Wait for `CREATE_COMPLETE` (~3 min)

2.3 — Paste the stack outputs back into Refine

Open the **Outputs** tab of the new stack. Copy these four values to the Account Setup modal in Refine:

CF Output	Refine field
RoleArn	Role ARN
CostDataBucketName	Cost data bucket name
RefineServiceAccessKeyId	Service Access Key ID
RefineServiceSecretAccessKey	Service Secret Access Key (shown ONCE in the CF Outputs — copy now)

The Stack Name field auto-fills if you used the default.

Click **Save Outputs**, then **Validate**.

Optional: keyless access (no access key at all)

If your Refine server runs on **EC2 in the same AWS partition** as the account you're onboarding, you can skip the access key entirely. Instead of creating an IAM user, the monitored account trusts your Refine server's own IAM role directly.

Why prefer it: the `RefineServiceSecretAccessKey` output stays readable in the CloudFormation **Outputs** tab to anyone with `cloudformation:DescribeStacks`, for the life of the stack. In keyless mode no IAM user, no access key, and no secret output are ever created, so there is nothing to leak or rotate.

To use it:

1. In Refine, open the host stack's CloudFormation **Outputs** and copy `RefineHostRoleArn` (the IAM role your Refine EC2 instance runs as).
2. Deploy (or update) the account-onboarding stack with the `RefineHostRoleArn` parameter set to that value.
3. In the Refine Account Setup modal, set **Access Method** to *Keyless — this Refine server's IAM role*. The key fields disappear.
4. Click **Save Outputs**, then **Validate**.

The stack's `CredentialMode` output tells you which mode it deployed in.

When you cannot use keyless — leave `RefineHostRoleArn` blank and use the access key method if Refine runs outside AWS (Azure VM, GCP, on-prem, or a laptop), or if Refine is in the commercial partition and the monitored account is in GovCloud. IAM principals cannot cross the commercial/GovCloud boundary, so those setups require a key.

If validation returns a green ✓, the account is live. The first nightly Lambda collection happens automatically; or click **Sync Now** to trigger one now.

First-day note: cost data takes up to 24 hours for the first AWS Cost & Usage Report to drop into the bucket. Inventory data (EC2, RDS, S3, etc.) appears within minutes of the first sync. The cost number on the dashboard will be \$0.00 until the first CUR drops — that's normal.

Repeat 2.1–2.3 for every AWS account you want monitored.

Part 3 — Connect your Azure subscription

Azure onboarding is faster than AWS — one Cloud Shell command provisions everything.

3.1 — Open Azure Cloud Shell

Go to <https://shell.azure.com> in the **tenant that owns the subscription** you want to monitor. Pick **Bash** mode.

Confirm you're in the right subscription:

```
az account show --query '[name, id]' -o tsv
az account set --subscription <your-subscription-id>    # if you need to switch
```

3.2 — Upload the onboarding script

In Cloud Shell, click **Manage files** → **Upload** (the icon in the top toolbar). Pick the file from your delivery ZIP at:

```
infra/azure/refine-onboarding.sh
```

After upload completes, in the shell:

```
chmod +x refine-onboarding.sh
./refine-onboarding.sh
```

Type **y** at the confirmation prompt.

The script takes ~60 seconds. It creates:

- An Entra ID **App Registration** + **Service Principal** (named `refine-readonly-<random>`)
- A 1-year **Client Secret** for that Service Principal
- A **Reader** role assignment at subscription scope
- A **Storage Account** + Blob container (`refine-cost-export`)
- A daily **Cost Management Export** dropping CSV into that container

If Cost Management Export creation is skipped (Pay-As-You-Go subscriptions require EA / MCA for Cost Management), don't worry — inventory still works fine. You can wire up the export manually later in Portal → Cost Management → Exports.

3.3 — Copy the seven values

At the end, the script prints:

```
Azure Subscription ID : ed59e273-...
Azure Tenant ID       : c636ad17-...
Azure Client ID       : a4d93cb1-...
Azure Client Secret   : ABC8Q~xxxx...
Azure Environment     : Public
Storage Account       : refinecost4b0c5c55
Storage Container     : refine-cost-export
```

Copy the Client Secret immediately — Azure never shows it again. If you lose it, just re-run the script; it generates a fresh App Registration each time (old ones can be cleaned up later in Entra ID → App Registrations).

3.4 — Paste into Refine

In Refine: **Cloud Accounts** → **Add Account** → **Azure**. Fill in all five required fields (Subscription ID, Tenant ID, Client ID, Client Secret, Environment) plus the optional Account Name. Click **Add Account**.

Validation runs automatically; you should see a green ✓ within a few seconds.

Click **Sync Now** to pull inventory right away. Inventory lands in seconds; cost data lands after the first daily export drop (≤ 6 hours).

Part 4 — Daily use

The first time inventory lands you'll have data on the following pages:

4.1 — Dashboard (/)

Top of the page: **Spend by Cloud** widget showing the AWS / Azure split. Below: nine resource category tiles (Compute, Databases, Storage, Block Storage, Serverless, Containers, Network, Caching, Logs). Each tile shows per-cloud counts (e.g. "AWS 8 · Azure 3"). Click any tile to drill into the unified inventory page for that category.

4.2 — Cost Cleanup (/cost-cleanup)

The "low-hanging fruit" page. Items here are 1-click wins — orphaned NAT Gateways, idle Public IPs, gp2 EBS volumes that should be gp3, S3 buckets without lifecycle policies, oversized Azure VMs. Sorted by savings descending. Each row has a Copy CLI button that gives you the exact AWS CLI / Azure CLI / gcloud command to execute the change.

This is the page to check weekly.

4.3 — Recommendations (/recommendations)

The fuller, more nuanced list. Rightsizing recommendations based on 1-month, 3-month, and 12-month utilization metrics. Use the period toggle at the top of the page to switch between windows — sizing based on 1-year data is typically the safest.

4.4 — Inventory pages (/inventory/{category})

One page per resource category, unified across clouds. Filter by cloud at the top (All / AWS / Azure). Filter by region, recommendation type, or free-text search. Click **Export CSV** to dump the current filtered view for offline analysis or reporting.

4.5 — Savings Tracking (/savings-report)

Once you start acting on recommendations, this page shows actual savings realized vs the baseline (when each resource was first observed). The Refine attribution engine matches "instance X was downsized from m5.xlarge to m5.large on date

Y" to a delta in your billed cost.

4.6 — Sharing an account with a teammate

Each account row has a **Share** button. Click it to download a JSON config file containing everything the teammate needs to connect to the same account from their Refine — except secrets.

You'll need to share the secret (AWS Secret Access Key / Azure Client Secret) via a **secure channel** (1Password, encrypted email, in-person). The JSON file alone is safe to email.

The teammate then uses **Cloud Accounts** → **Import Account** in their Refine to paste the JSON + the secret you sent them separately.

Part 5 — Day-2 operations: the `refine.sh` script

Every customer delivery ZIP ships a `refine.sh` script at the top level for common operational tasks. Run it from `/opt/refine`:

Command	What it does
<code>./refine.sh status</code>	Show container + image status
<code>./refine.sh start</code>	Start the stack
<code>./refine.sh stop</code>	Stop the stack
<code>./refine.sh restart</code>	Restart
<code>./refine.sh update</code>	Pull the latest image and recreate the container
<code>./refine.sh logs</code>	Last 200 lines of logs
<code>./refine.sh logs -f</code>	Follow logs live (Ctrl+C to stop)
<code>./refine.sh shell</code>	Open a shell inside the container
<code>./refine.sh backup</code>	Snapshot the data dir to <code>backups/</code>
<code>./refine.sh version</code>	Print the running Refine version
<code>./refine.sh health</code>	Hit <code>/health</code> locally
<code>./refine.sh url</code>	Print the URL to open in a browser

Auto-updates: the container has a Watchtower sidecar (`refine-watchtower`) that checks `ghcr.io` for a new version once an hour and restarts Refine on it. When a new version is available, a banner in the Refine UI also offers an **Update Now** button, which triggers Watchtower immediately — no command line required.

To confirm auto-update is healthy:

```
sudo docker logs refine-watchtower 2>&1 | head -8
```

You should see `Watchtower 2.3.2` and `Scheduling first run`, and **no** `client version ... is too old` or `Periodic runs are not enabled` lines. If you see either, your `docker-compose.yml` predates the fix — contact support@blacktip-ops.com for the updated file.

Manual updates: `./refine.sh update` from the shell does the same thing on demand. Equivalent to `docker compose pull && docker compose up -d`.

Backups before risky changes:

```
./refine.sh backup          # before
# ... do the thing ...
# if you need to roll back:
./refine.sh stop
tar -xzf backups/refine-data-<timestamp>.tar.gz
./refine.sh start
```

The `data/` directory holds your SQLite DB and license file. Backing it up is sufficient to recover the entire installation.

Part 6 — Troubleshooting

Symptom	Diagnosis	Fix
<code>./refine.sh status</code> says container is "Restarting" or "Exited"	Most common cause is OOM on <code>t3.micro</code>	Bump instance to <code>t3.small</code> or <code>t3.medium</code> . Stop EC2 → Change instance type → Start.
<code>./refine.sh logs</code> shows "license file not found"	The CloudFormation template couldn't download the ZIP, or <code>refine.license</code> is missing from it	Verify the ZIP is at <code>s3://<bucket>/refine-delivery.zip</code> (exact name), and that the ZIP has a <code>refine.license</code> at its root
Login page loads but Add Account → AWS validation fails with "Access Denied"	The Refine service IAM user can't assume the role	Re-check that <code>RefineServiceAccessKeyId</code> + <code>RefineServiceSecretAccessKey</code> from the CF Outputs are pasted correctly. They're shown only once in CF Outputs — if you missed them, re-deploy the stack (or use the CF console to view the new secret in the IAM user's <code>AccessKey1</code> if your account allows it)
Add Account → Azure validation fails: "Azure SDK not installed"	You're on an older Refine image that predates Azure SDK support	<code>./refine.sh update</code> to pull the latest image
Add Account → Azure validation fails: "AAD authentication failed"	Tenant ID / Client ID / Client Secret mismatch, or you're in the wrong Azure environment	Re-check all three against the Cloud Shell script output. Confirm <code>Environment = Public</code> (not USGov / China)
Validation passes for Azure but no resources appear in inventory	First sync hasn't run yet	Click Sync Now on the Cloud Accounts page. Wait 30–60 seconds, then refresh inventory page
EC2 recommendations disappear after the nightly sync and reappear after a daytime Sync Now	Fixed in v3.14.6: instances stopped at the 02:00 UTC sync were skipped. Every instance that ran during the month is analyzed now	Update Refine
Windows or SQL Server instances show a higher cost than before	Refine now prices the platform AWS reports (Linux rate + licence per vCPU-hour); earlier versions priced everything as Linux, 30-45% low	Nothing to do — the new figures are the on-demand list rates. Bring-your-own-licence AMIs report a plain platform and are priced as such
Savings Tracking shows negative resource savings, or <i>New Resources Added</i> jumps, for Windows or SQL Server instances after updating to v3.14.5 or later	The account's onboarding stack froze each resource's baseline at the Linux price (template v2.34.0 or older) and the current cost now includes the licence portion	Update the account-onboarding stack to template v2.35.0 or later, then click Sync Now . The next collector run re-bases the baseline of every re-priced instance to the new price basis, so only real changes count
Savings Plans page shows "month to date" instead of the last 30 days, no	The account's onboarding CloudFormation stack predates the daily	Update the account-onboarding stack to the latest template, then click Sync Now . Until then Savings Plan sizing uses the

Symptom	Diagnosis	Fix
utilization-by-day bars, or the Commitment Breakdown says the stack "does not collect hour-by-hour running data yet"	commitment series and the hourly running data (template v2.34.0)	conservative per-instance rule and cannot see staggered schedules
Savings Plans or Reserved Instances you bought don't appear in the Savings Report	The account's onboarding CloudFormation stack predates commitment collection (template v2.33.0)	Update the account-onboarding stack to the latest template, then click Sync Now . Up to 12 months of Savings Plan and RI history is filled in on the first sync. The Savings Report explains in its Savings Plans card if a stack still needs updating.
Dashboard shows <code>\$0.00</code> total cost on day 1	Normal — AWS Cost & Usage Reports take 24 h to first drop, Azure Cost Management Export takes ≤ 6 h	Wait. Or for AWS, click Refresh Costs on the Cost Overview page after the first CUR drops
Browser shows "Connection refused" hitting <code>http://ip:8000</code>	Either container is down OR your IP is not in <code>AllowedAppCidr</code>	Run <code>./refine.sh status</code> to check the container. Then check your current IP at <code>curl ifconfig.me</code> and compare to <code>AllowedAppCidr</code> in the CloudFormation parameters. Update the SG inbound rule if your IP changed.
Sidebar says "Licensed until ..." but license validation fails on login	License file moved or permissions wrong	<code>ls -la /opt/refine/refine.license</code> should show it owned by the user running Docker. Re-extract from the delivery ZIP if needed.
You see Azure resources you don't recognize	Cost Management Export may include sub-resources from other deployments	Filter by Account in the inventory page to see only the subscription you expect. The unrecognized rows are likely from a hidden Resource Group; check Azure Portal to confirm
Refine is several versions behind and never updated itself, or Update Now fails with "Could not reach Watchtower" (event <code>SYS_003</code>)	The auto-update sidecar isn't running. Check <code>sudo docker logs refine-watchtower 2>&1 head -20</code>	<code>client version 1.25 is too old</code> or <code>Periodic runs are not enabled</code> : your <code>docker-compose.yml</code> predates the fix — contact support@blacktip-ops.com for the updated file, then <code>sudo ./refine.sh update</code> . <code>client version 1.44 is too new</code> : Docker Engine is older than 25 — <code>sudo ./refine.sh update</code> handles it. <code>no basic auth credentials</code> or <code>pull access denied</code> (the session line still says <code>Failed=0</code>): <code>sudo bash registry-login.sh</code> , then <code>sudo ./refine.sh update</code>
Watchtower auto-update happened but the new version "looks the same"	Browser cache	Hard refresh: Ctrl+Shift+R (Win/Linux) or Cmd+Shift+R (Mac). Or open in an incognito window
In-app banner says "Update available" but <code>./refine.sh update</code> says "already up to date"	The banner shows a banner for any tag newer than <code>/app/VERSION</code> ; sometimes a release is pushed but not yet pulled	<code>./refine.sh restart</code> to recycle the container so it re-checks
CloudFormation stack rolls back with <code>"The maximum</code>	Your AWS account has hit the default Elastic IP cap (5 per region)	See "Cannot launch the host stack — Elastic IP limit reached" below

Symptom	Diagnosis	Fix
number of addresses has been reached"		

System event codes

Every System Event Refine records carries one of these codes (the **System Events** page shows the full remediation steps for each; `GET /v1/system/events` returns them). The table is generated from the registry the app ships with (`scripts/gen_doc_tables.py`), so it matches the version you are running.

Code	Severity	What happened	First thing to do
SYNC_001	error	Lambda function not found — The data collector Lambda was not found in the customer AWS account.	Go to AWS Accounts and check the account status
SYNC_002	error	Access denied invoking Lambda — The Refine IAM role does not have permission to invoke the data collector Lambda.	Check the CloudFormation stack status in your AWS Console
SYNC_003	warning	No validated AWS accounts — No validated AWS accounts were found for this tenant.	Go to AWS Accounts and add an account
SYNC_004	error	S3 data read failed — Could not read inventory data from the customer S3 bucket.	Verify the S3 bucket exists in your AWS account
SYNC_005	error	Sync completed with account errors — One or more AWS accounts failed during data sync.	Go to AWS Accounts and check each account's status
SYNC_006	error	Lambda invocation timed out — The data collector Lambda took longer than Refine was willing to wait. This usually means the customer AWS account has a large inventory, or the Lambda is hitting AWS API throttling.	Re-run Sync Now — subsequent runs are often faster (partial data cached)
AWS_001	error	AWS credentials invalid or expired — The per-account service credentials (Access Key ID / Secret Access Key) are invalid or have been deactivated in AWS IAM.	Check the IAM user in your AWS Console — is the access key still active?
AWS_002	error	STS AssumeRole failed — Could not assume the Refine cross-account IAM role.	Verify the Role ARN in Refine matches the role in your AWS account
AWS_003	warning	Missing per-account credentials — This AWS account has no service credentials saved. Sync cannot proceed without Access Key ID and Secret Access Key.	Go to AWS Accounts → click the account → Details
AUTH_001	warning	Repeated authentication failures — Multiple failed login attempts detected from your IP address.	If this is unexpected, consider changing your password. Rate limiting is active — wait a minute before trying again.
LICENSE_001	warning	License expiring soon — Your Refine license is expiring within 30 days.	Contact Blacktip Solutions at support@blacktip-ops.com to renew. After renewal, replace <code>refine.license</code> and restart the container.
SYS_001	critical	Database migration failed — A database schema migration failed during startup.	Check container logs for the migration error: <code>docker logs refine</code>
SYS_002	error	Scheduler failed to start — The nightly sync scheduler could not start. Data will not sync automatically.	Restart the container: <code>docker compose down && docker compose up -d</code>
SYS_003	error	Auto-update sidecar not responding — Update Now could not reach the Watchtower sidecar, or it	Check the sidecar: <code>sudo docker logs refine-watchtower 2>&1 head</code>

Code	Severity	What happened	First thing to do
		rejected the request. If Watchtower is down, Refine is also not updating on its hourly schedule.	-20

Cannot launch the host stack — Elastic IP limit reached

If the CloudFormation deploy in Part 1.2 rolls back with:

```
Resource handler returned message: 'The maximum number of addresses has been reached.' (Service: Ec2, Status Code: 400, ...)
```

...it means your AWS account is at the default cap of **5 Elastic IPs per region**. Try these solutions in order — the first usually fixes it within a minute.

Solution 1 — Release unused EIPs (most common cause)

Orphaned EIPs (allocated but not associated with a running instance) still count against the cap and each cost ~\$3.60/month. List them:

```
aws ec2 describe-addresses --region us-east-1 \
  --query 'Addresses[?AssociationId==null].[PublicIp,AllocationId,Tags]' \
  --output table
```

For each EIP you no longer need, release it:

```
aws ec2 release-address --region us-east-1 --allocation-id eipalloc-XXXX
```

Then **retry the CloudFormation deploy**. This resolves the issue ~80% of the time.

Solution 2 — Request a quota increase

If all 5 EIPs are legitimately in use, raise the cap:

1. AWS Console → **Service Quotas** → AWS services → **Amazon EC2**
2. Search for "**EC2-VPC Elastic IPs**"
3. Click **Request quota increase** and ask for **10** (or **20**)
4. Approval is typically automatic and arrives in 5–30 minutes

Then retry the deploy.

Solution 3 — Deploy without an Elastic IP

If you don't need a stable public IP (you'll only access Refine via VPN / SSM / private DNS), edit `infra/refine-server-auto.yaml` in your delivery ZIP and **comment out** the `RefineEIP` and `RefineEIPAssociation` resources plus any `!Ref` `RefineEIP` references in the Outputs section. Re-upload the edited template and retry the stack.

You'll connect to Refine via the EC2's auto-assigned public IPv4 (which changes on stop/start) or, better, via SSM Session Manager + port forwarding:

```
aws ssm start-session --target i-xxxxxxx \
  --document-name AWS-StartPortForwardingSession \
  --parameters '{"portNumber":["8000"],"localPortNumber":["8000"]}'
# then open http://localhost:8000
```

Solution 4 — Use a different AWS region

The EIP cap is **per region**. If `us-east-1` is full but `us-west-2` is empty, deploy the Refine host stack in `us-west-2` and use cross-region calls for cost/inventory collection. Latency from a `us-west-2` host to `us-east-1` monitored resources is fine (~70 ms per API call; sync runs nightly).

Tip: *Solution 1 is almost always the right answer. Run the `describe-addresses` command first — most customers find at least one orphan EIP from an old experiment.*

Getting a fresh log dump for Blacktip support

If you contact us, please include:

```
cd /opt/refine
./refine.sh version
./refine.sh logs > /tmp/refine-logs.txt
sudo docker logs --tail 500 refine-watchtower > /tmp/watchtower-logs.txt 2>&1
tar -czf /tmp/refine-diag.tar.gz /tmp/refine-logs.txt /tmp/watchtower-logs.txt
```

Attach `/tmp/refine-diag.tar.gz` to your email. **No license file, no credentials, no customer data — just diagnostic logs.**

Part 7 — Removing an account

Click **Remove** on a cloud account row. Confirm in the dialog.

This is a **real hard delete**:

- The account row is physically removed from the database
- Every inventory + cost + recommendation row referencing it is also deleted
- The toast shows the cascade count (e.g. "1,247 associated inventory rows")

The CloudFormation stack / Azure Service Principal in the cloud is **not** touched. To clean those up:

- **AWS:** `aws cloudformation delete-stack --stack-name <name>` in the monitored account
- **Azure:** `az ad app delete --id <client-id>` in the Cloud Shell where you ran the onboarding script

If you change your mind, you can re-add the account by repeating Part 2 or 3.

Part 8 — Need help?

Email support@blacktip-ops.com with:

- Your customer ID (top-left of the sidebar)
- Your Refine version (`./refine.sh version`)
- A short description of what you tried + what happened
- The diagnostic tar from the Troubleshooting section above

Response time: **24 hours for non-blocking, 2 hours for production-down.**

Appendix — Quick reference

URLs you'll use a lot

Page	URL on your server
Dashboard	<code>http://<ip>:8000/</code>
Cost Cleanup (start here weekly)	<code>http://<ip>:8000/cost-cleanup</code>
Recommendations	<code>http://<ip>:8000/recommendations</code>
Compute inventory (EC2 + Azure VMs)	<code>http://<ip>:8000/inventory/compute</code>
Databases	<code>http://<ip>:8000/inventory/databases</code>
Cloud Accounts (Add / Remove / Share)	<code>http://<ip>:8000/accounts</code>
Savings Tracking	<code>http://<ip>:8000/savings-report</code>
Usage Trends	<code>http://<ip>:8000/trends</code>

Where things live on the EC2

```
/opt/refine/
├─ refine.license      ← your license file, do not delete
├─ docker-compose.yml  ← stack definition
├─ refine.sh           ← ops script
├─ data/               ← SQLite DB + collected data (BACK THIS UP)
│   └─ refine.db
├─ backups/            ← snapshots from ./refine.sh backup
└─ .env                ← environment overrides (port, etc.)
```

Common AWS CLI snippets

```
# What's my current public IP (for AllowedAppCidr updates)?
curl ifconfig.me

# Update Refine SG to allow a new /32
aws ec2 authorize-security-group-ingress \
  --group-id sg-XXXXXX \
  --protocol tcp --port 8000 --cidr 1.2.3.4/32

# Tail Refine logs via SSM (no SSH key needed)
aws ssm start-session --target i-xxxxxxx
# then inside: cd /opt/refine && ./refine.sh logs -f
```

Common Azure CLI snippets

```
# What App Registrations do I have for Refine?
az ad app list --display-name "refine-readonly-*" --query "[].{name:displayName,id:appId}"

# Rotate the client secret
APP_ID=$(az ad app list --display-name "refine-readonly-XXXXXXX" --query "[0].appId" -o tsv)
az ad app credential reset --id $APP_ID --years 1

# Then paste the new password into Refine: Cloud Accounts → Edit → Azure Client Secret
```

That's the whole loop. Most weeks you'll only open the Cost Cleanup page, glance at recommendations, and otherwise leave Refine alone. Reach out if anything in this guide is unclear or stale.